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Gegeven is de breuk $z = \dfrac{\sqrt{2}}{1+i}$. We zoeken de goniometrische vorm door eerst teller en noemer afzonderlijk om te zetten.

Teller $z_1 = \sqrt{2}$:

$r_1 = \sqrt{(\sqrt{2})^2 + 0^2} = \sqrt{2}$

$\alpha_1 = 0 \Rightarrow z_1 = \sqrt{2}\operatorname{cis} 0$

Noemer $z_2 = 1 + i$:

$r_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\tan \alpha_2 = \dfrac{1}{1} = 1 \Rightarrow \alpha_2 = \dfrac{\pi}{4} \Rightarrow z_2 = \sqrt{2}\operatorname{cis}\dfrac{\pi}{4}$

Quotient bepalen:

$$z = \frac{z_1}{z_2} = \frac{\sqrt{2}\operatorname{cis} 0}{\sqrt{2}\operatorname{cis}\frac{\pi}{4}}$$

$$z = \frac{\sqrt{2}}{\sqrt{2}}\operatorname{cis}\left(0 - \frac{\pi}{4}\right)$$

$$z = 1\operatorname{cis}\left(-\frac{\pi}{4}\right)$$

Antwoord: $z = \operatorname{cis}\left(-\dfrac{\pi}{4}\right)$

References